Task 1: Data Cleaning & Preprocessing

Detailed Description

This foundational task involves transforming raw, unstructured datasets (e.g., sales records, user logs, survey responses) into analysis-ready formats. Real-world data often contains inconsistencies, gaps, and errors that distort insights. Interns will learn to diagnose data health issues and apply systematic cleaning techniques to ensure reliability for downstream analysis.

Key Responsibilities

Step	Actions	Tools/Methods
1. Data Ingestion	- Load datasets from CSV/Excel files, databases, or APIs - Perform initial sanity checks (shape, basic stats)	`pd.read_csv()`, `pd.read_sql()`, `df.head()`, `df.info()`
2. Deduplication	- Identify and remove exact/partial duplicate rows - Detect cross-column redundant entries	`df.duplicated()`, `df.drop_duplicates()`, `df[col].unique()`
3. Column Management	- Drop irrelevant columns (e.g., IDs, free-text notes) - Rename columns for clarity - Reorder columns logically	`df.drop(columns=[])`, `df.rename()`, `df.reindex()`

4. Missing Value Handling	- Diagnose missingness patterns (MCAR, MAR, MNAR) - Apply strategies: • *Deletion*: Drop rows/columns if >70% missing • *Imputation*: Fill with mean/median (numerical), mode (categorical), or interpolate (time-series)	`df.isna().sum()`, `df.dropna()`, `df.fillna()`, `SimpleImputer()`
5. Data Type Correction	- Convert strings to datetime objects (e.g., `"2023-12-01"` → `datetime64`) - Fix numerical columns trapped as strings (e.g., `"1,000"` → `1000`) - Standardize categorical encoding	`pd.to_datetime()`, `pd.to_numeric()`, `astype()`, `str.replace()`
6. Format Standardization	- Normalize text (lowercase, strip whitespace) - Fix inconsistent units (e.g., "kg" vs "lbs") - Map categorical variants (e.g., "M"/"Male" → "Male")	`str.lower()`, `str.strip()`, `np.where()`, `df.replace()`

Skills Gained

- ✅ Advanced `pandas` Operations: Mastery of data manipulation using DataFrames/Series.
- ✅ Data Quality Assessment: Ability to quantify and visualize data issues (e.g., missingness matrix, inconsistency reports).
- ✅ Strategic Decision-Making: Judgement in choosing between imputation/deletion based on data context.
- ✅ Real-World Problem Solving: Experience handling ambiguous, unstructured datasets.
- ✅ Efficiency Optimization: Vectorized operations for large datasets.

Example Workflow

```
import pandas as pd

# Load data
df = pd.read_csv("messy_sales_data.csv")

# Remove duplicates & irrelevant columns
df = df.drop_duplicates().drop(columns=["notes", "temp_id"])

# Handle missing values
df["revenue"].fillna(df["revenue"].median(), inplace=True)
df.dropna(subset=["customer_id"], inplace=True)

# Convert data types
df["sale_date"] = pd.to_datetime(df["sale_date"], errors="coerce")
df["price"] = pd.to_numeric(df["price"].str.replace("$", ""))

# Standardize formats
df["product"] = df["product"].str.lower().str.strip()
df["region"] = df["region"].replace({"west": "Western", "SOUTH": "Southern"})
```

Outcome: A clean, analysis-ready dataset stored as `cleaned_sales_data.csv`.

Pro Tip: Always validate changes with `assert df[col].isna().sum() == 0` or `assert df.duplicated().sum() == 0` to prevent silent failures.

Here are curated datasets for practicing Task 1: Data Cleaning & Preprocessing, along with download links and key characteristics:

🧹 Beginner-Friendly Datasets

1. Customers Dataset
    - Description: Simulated e-commerce customer records with duplicates, missing IDs, inconsistent formatting (names, dates, regions), and invalid entries.
    - Size: 100–2M records.
    - Download: customers-100.csv [3].
2. People Dataset
    - Description: Demographic data with mixed date formats, inconsistent job titles, missing values, and malformed phone numbers.
    - Size: 100–2M records.
    - Download: people-100.csv [3].
3. Organizations Dataset
    - Description: Company records with inconsistent industry categories, missing employee counts, and website formatting errors.
    - Size: 100–2M records.
    - Download: organizations-100.csv [3].

📊 Intermediate Datasets

4. Retail Sales Data
    - Description: Sales transactions with date parsing issues, outliers in sales/quantity, and store/item ID inconsistencies.
    - Download: train.csv (Kaggle-sourced) [9].
5. Spotify Streaming History

- Description: 149k+ music logs with timestamp errors, missing track IDs, and inconsistent artist names.
- Download: Spotify Dataset (Maven Analytics) [5].

⚙️ Advanced Datasets

6. Data Science Job Postings
   - Description: Glassdoor scrapings with salary outliers, unstructured text (job descriptions), and mixed location formats.
   - Download: Glassdoor Job Data [10].
7. NYC Taxi Trips
   - Description: 200M+ rows with GPS coordinate errors, fare discrepancies, and missing passenger counts.
   - Download: NYC Taxi Data [11].